

Correlation of Scattering Loss, Sidewall Roughness and Waveguide Width in Silicon-on-Insulator (SOI) Ridge Waveguides

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Abstract—We use star couplers to measure the relative scattering losses of silicon-on-insulator (SOI) ridge waveguides of various widths over the range of 1.75 to 0.2 μm in a single measurement. The scattering loss data obtained for waveguides fabricated by different photolithography and e-beam base processes correlate well with the measured root-mean-square roughness of the waveguide sidewalls obtained using SEM image analysis, and are in qualitative agreement with the prediction of simple scattering loss theory.

Index Terms—Scattering loss, sidewall roughness, silicon-on-insulator (SOI) waveguide, star coupler.

I. INTRODUCTION

OPTICAL scattering loss is a major problem with high-index-contrast silicon-on-insulator (SOI) waveguides. Scattering loss is caused by sidewall roughness at the interfaces between waveguide core and cladding [1]–[3]. For example, when the roughness exceeds 10 nm (root-mean-square value) for a 0.5 μm wide Si channel waveguide, the resultant attenuation loss can rise above 30 dB/cm [4].

SOI waveguide scattering loss increases drastically with decreasing waveguide width due to a stronger interaction of the guided modes with the waveguide sidewall [4]. In addition, scattering loss of small SOI waveguides varies with waveguide mode polarization state because of the differences in the effective index and profile of the guided modes. This results in a measurable polarization dependent loss (PDL), which is undesirable in waveguides coupled to any optical fiber with an unknown state of polarization. Minimizing scattering loss is

thus a primary consideration in optimizing the fabrication of silicon photonic components.

Various studies intended to model the effect of roughness and waveguide dimensions on scattering loss have been reported in the literature [5]–[8]. The Payne-Lacey (PL) model [7] is a commonly accepted analysis that uses a closed-form analytical expression to describe the scattering loss in two-dimensional planar (or slab) waveguides as a function of the fundamental waveguide parameters, and the root-mean-square (r.m.s) roughness (σ) and correlation length (L_c). The scattering loss is given as

$$\alpha = 4.34 \frac{\sigma^2}{\sqrt{2} k_o d^4 n_1} g \cdot f \quad (1)$$

where α is the waveguide scattering loss in dB per unit length, k_o , d and n_1 are the free-space wave vector, the waveguide half width, and the core index, respectively. The function g is determined purely by the waveguide geometry, while f is determined by the correlation length and index step of the waveguide [7].

This expression predicts that the scattering loss increases with the mean square deviation of the surface roughness, σ^2 , and, for any given roughness, the scattering loss increases rapidly as waveguide dimensions are reduced.

Our aim is to develop an experimental approach to measure waveguide scattering loss and sidewall roughness statistics, establish their correlation, and compare experimental results with the PL model predictions. Such correlations will enable process engineers to improve fabrication processes to reduce roughness and scattering. In order to facilitate this study, a waveguide loss measurement technique is needed to provide enough data that meaningful correlations can be established.

The existing loss measurement techniques, namely the cut-back method and Fabry-Perot (FP) resonance method [9], are not well suited for comparing large numbers of waveguides. Both methods can be very time consuming and the results depend critically on the quality and consistency of the waveguide facets. Measurement-to-measurement variations in optical alignment can often limit accuracy of the cut-back method [9]. The accuracy of the FP method depends on having a precise value of the facet reflectivity. This reflectivity is difficult to determine, because optical facets defining the cavity may have rough texture and profile, which introduces unpredictable reflection losses [10]. Furthermore, the facet reflectivity for small waveguides may differ significantly from the Fresnel formulae, which are based on a single plane wave calculation.

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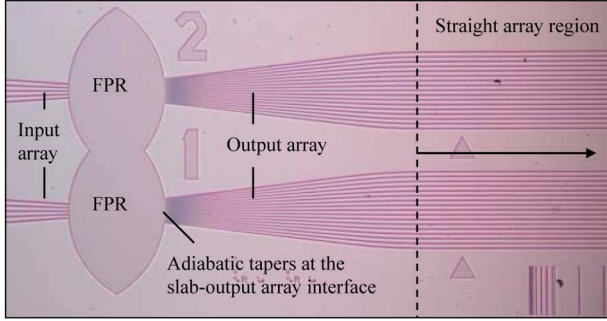


Fig. 1. Optical image of two fabricated 5×17 star couplers for waveguide scattering loss measurement.

For instance, the calculated facet reflectivity of III-V cavity lasers can vary from 20% to 40% depending on waveguide thickness, index contrast and optical polarization [11]. Apart from the uncertainty in the reflectivity value, interference from leaky modes and stray light can cause variations in the peak height of the F-P fringes and imposes difficulty in determining an accurate fringe contrast [12].

To overcome these limitations, a non-destructive technique using SOI star couplers to measure excess scattering induced waveguide loss has been devised in this study. The accuracy of this technique is independent of the input coupling efficiency and facet reflectivity. The method measures the relative waveguide loss of many different waveguides at various waveguide widths in a single step. By measuring these losses on various SOI star couplers fabricated from different processes, this technique provides reliable feedback to process development and optimization.

II. OPERATING PRINCIPLE AND DESIGN

A. Operating Principle

The star coupler used for scattering loss measurement has an input array of five waveguides and an output array of seventeen waveguides, connected by a free-space propagation slab region (FPR), as shown in Fig. 1.

In this star coupler, a traveling Gaussian wave from any input waveguide will be diffracted at the input array-slab interface and project a Fraunhofer far-field profile to illuminate all output waveguides at the slab-output array interface. The Fraunhofer profile assumes an approximately Gaussian output pattern, when the input waveguide profile is approximately Gaussian. Each waveguide in the array will propagate light to the output facet. In the absence of optical loss, the intensity distribution across the output array at the facet should have the same profile as was launched at the slab-output array interface. If extra scattering loss is introduced into any waveguide of the output array, a drop in the intensity of the corresponding waveguide output will be observed, as shown in Fig. 2. Since scattering loss increases rapidly with the reduction in waveguide width, star coupler with an array of output waveguides appears to be an excellent device to study the loss at different widths.

Input waveguides of our star coupler have identical width of $3 \mu\text{m}$ but output waveguides have varying widths. The width of all odd output waveguides is fixed at $2 \mu\text{m}$, while the width of

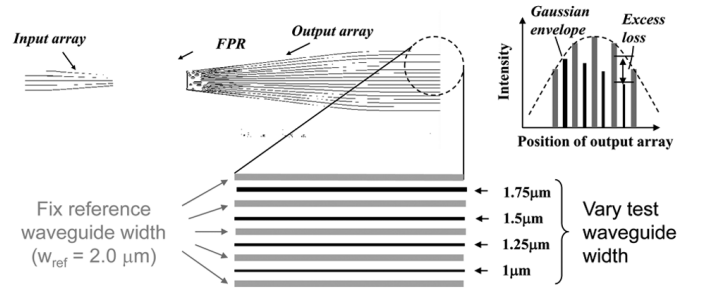


Fig. 2. Schematic illustration of waveguide scattering loss measurement using a star coupler. The odd waveguides are reference waveguides with fixed width of $2 \mu\text{m}$, while even waveguides are test waveguides of varying width. The output intensity at the far field contains multiple peaks with a Gaussian envelope, as shown in the upper right corner.

even waveguides varies from 0.2, 0.4, 0.6, 0.8, 1, 1.25, 1.5 to $1.75 \mu\text{m}$. By fixing the width of every odd waveguide in the array to be $2 \mu\text{m}$ while varying the width of remaining even waveguides, lower peak intensities will be seen for the even waveguides as a result of the increased scattering due to width reduction. Using the odd waveguides as reference waveguides with intensities following the symmetric far-field envelope, as illustrated in Fig. 2, the extra scattering loss of any even waveguide can then be calculated from the ratio of its measured output intensity to the average intensity of the two adjacent reference waveguides. This excess loss represents the relative scattering loss of the even waveguides, with respect to the reference waveguide

$$\alpha = \log_{10} \left(\frac{I_n}{0.5(I_{n-1} + I_{n+1})} \right), \quad n = 2, 4, 6, \dots, 16 \quad (2)$$

where I_n is the measured intensity of the n th waveguide (n is an even number).

The measurement of this excess loss can be done relatively easily using an infra-red (IR) camera to capture the near field optical image of all output waveguides in the array. Output intensity of each waveguide can be measured from the image, from which the excess loss can be determined. Since one image contains the intensity information of all waveguides, the relative scattering loss of all waveguides can thus be measured in a single step.

B. Design

A $10 \mu\text{m}$ -wide gap between input waveguides is designed to avoid mode coupling. The radius of curvature of both array-slab interfaces is carefully calculated to be $400 \mu\text{m}$, so that the curvature at the output array-slab interface matches the wavefront of the diffracted light from any input waveguide, after propagating through the slab region. At the output array-slab interface, the width of output waveguides is tapered to $5 \mu\text{m}$ to receive maximum power and to minimize back reflection from the curved wall of the interface to the FPR, as shown in Fig. 3(a) and (b).

This leaves a minimum gap of $1 \mu\text{m}$ in between adjacent output waveguides. The output waveguides are adiabatically tapered down to their respective widths over a length of $350 \mu\text{m}$ and converge to form a straight and parallel array region towards the output facet. This 7-mm long straight array region is where

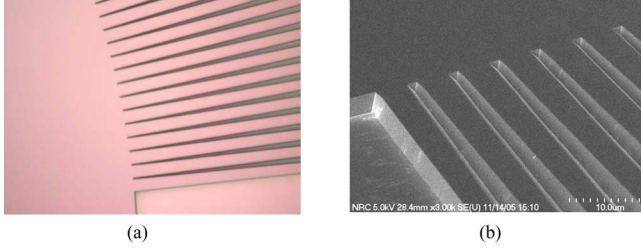


Fig. 3. (a) Top-view optical image of the slab-output array interface of our fabricated star coupler, (b) side-view SEM image showing the tapering of output waveguides at the same interface.

the effect of scattering is measured as a function of waveguide width, as shown in Fig. 1. Since the length of the tapered region is much less than the length of the straight array region, the total waveguide loss is dominated by the scattering at the straight region.

III. EXPERIMENT

A. Fabrication

Waveguide sidewall roughness may originate from the line edge corrugations of photoresist that is transferred to the waveguides during the subsequent etching, or from the etching process itself. Choosing the right combination of lithography, photoresist and dry etching process is therefore crucial in minimizing roughness. In this study, we have chosen to compare three processes using two different lithography tools commonly used to pattern SOI waveguides: a contact aligner and an e-beam direct writer. For the e-beam direct writing, we have studied both positive and negative resist for roughness comparison. An inductively coupled plasma (ICP) etcher has been used to etch the SOI waveguides given the excellent Si etch rate and good control of plasma energy. The three different fabrication processes to study waveguide roughness are shown in Fig. 4. These processes are mainly differ by their pattern transfer techniques and resist chemistry as follows.

- i) Process I: Contact lithography with a positive photoresist;
- ii) Process II: E-beam lithography with a positive e-beam resist and lift-off process;
- iii) Process III: E-beam lithography with a negative e-beam resist on a Cr hardmask layer.

The starting substrate was a (100) p-type SOI wafer (background doping concentration of $10^{15}/\text{cm}^3$) with a $2.5\ \mu\text{m}$ top Si layer on a $0.4\ \mu\text{m}$ buried SiO_2 . As illustrated in Fig. 4(a), one set of samples was patterned using contact lithography and dry etching in an ICP etcher with $\text{SF}_6:\text{C}_4\text{F}_8$ chemistry. UV light at $\lambda = 365\ \text{nm}$ was used to expose the SPR510 photoresist by contact lithography to achieve fine patterning of both input and output waveguide arrays. In order to minimize bend loss and polarization dependent coupling loss of the star coupler [9], [13], etch depth of the ridge waveguides was targeted at $1.8\ \mu\text{m}$.

Two other sets of samples using resists of opposite polarity were patterned by electron-beam direct writing. One set was patterned using a positive e-beam resist ZEP520, as shown in Fig. 4(b). After resist development, a $50\ \text{nm}$ thick Al layer was

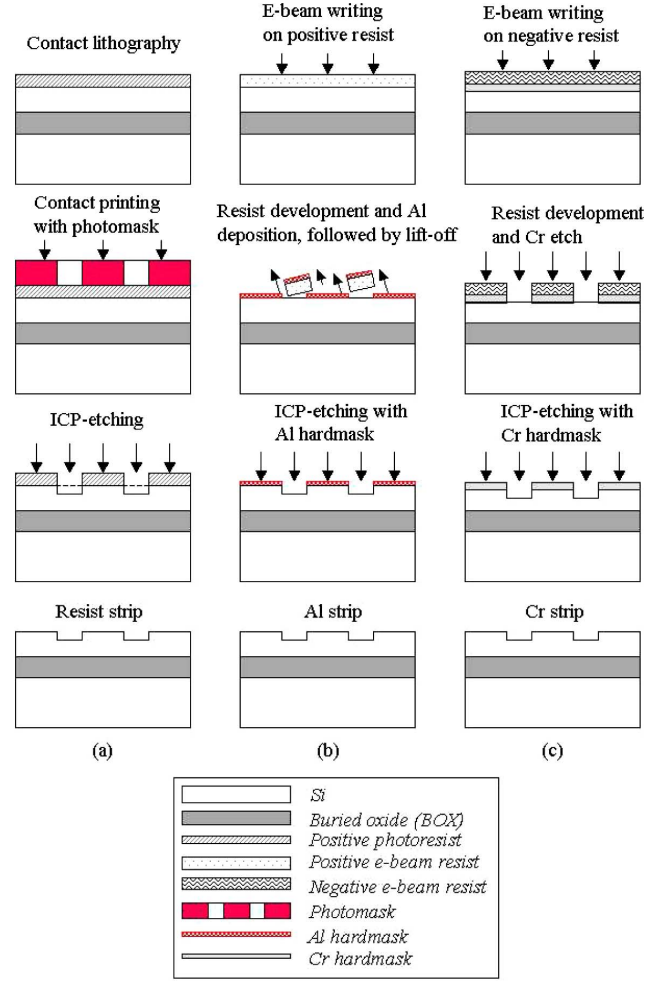


Fig. 4. Schematics of three different processes to fabricate SOI star couplers and waveguides with different sidewall roughness, (a) Process I: contact lithography, (b) process II: e-beam lithography and lift-off, (c) process III: e-beam lithography and Cr hardmask.

evaporated on this set of samples and the samples were then subjected to a lift-off treatment. Using the remaining Al layer as a hardmask, the samples were then dry etched in ICP etcher with the same $\text{SF}_6:\text{C}_4\text{F}_8$ recipe. The final set of samples was patterned using NEB22, a negative e-beam resist. A $50\ \text{nm}$ thick Cr layer was previously sputtered onto the samples serving as the hardmask for the dry etching of SOI waveguides in ICP etcher, as shown in Fig. 4(c). The Cr hardmask was etched in a $\text{Cl}_2:\text{O}_2$ plasma to facilitate the formation of a volatile etch product CrO_2Cl_2 with a low boiling point. The proportion of O_2 in the gas mixture was intentionally kept low to suppress the formation of $\text{Cr}(\text{CO})_6$, another unwanted by-product. An optimized Cr etch rate of $15\ \text{nm}/\text{min}$ was obtained using a combination of $50\ \text{sccm}\ \text{Cl}_2$ and $4\ \text{sccm}\ \text{O}_2$ in our ICP etcher, at room temperature with a chamber pressure of $30\ \text{mTorr}$ and DC power of $25\ \text{W}$. After the Cr etch, the samples were subsequently etched in $\text{SF}_6:\text{C}_4\text{F}_8$ plasma to form the ridge waveguides. The high selectivity of Cr to Si ($> 200 : 1$) gave an excellent process window for this etching step. The remaining Cr hardmask was then removed in a solution containing perchloric acid and ammonium cerium nitride, a standard wet etch agent for Cr. A

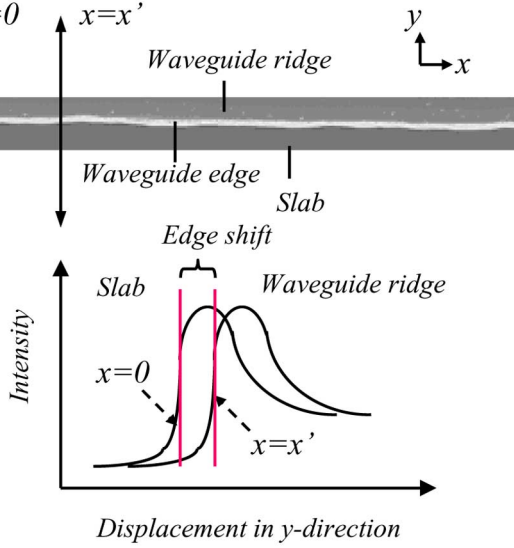


Fig. 5. An illustration of the sidewall roughness analysis by SEM. A shift in the intensity profile along the cross-section of the waveguide edge at the location where $x = x'$ is shown in the graph.

$0.5 \mu\text{m}$ thick SiO_2 film was subsequently deposited on all samples by plasma-enhanced chemical vapor deposition (PECVD) as an upper cladding layer to protect their structural integrity during facet polishing.

Since all three sets of samples were patterned with different resists and lithography techniques, the resulting sidewall roughness of the dry-etched SOI ridge waveguides of each sample set is different, depending mainly on the initial sidewall roughness and profiles of the patterned hardmask or resist. This provides a comparison for the investigation of scattering loss caused by different sidewall roughness.

B. Measurements

Roughness and Correlation Length Measurement: SEM analysis was employed to extract roughness characteristics of waveguides fabricated by all three processes. A series of continuous high-resolution top view SEM images of the waveguide edges at high magnification (at least $80 \times$) were collected. These collected images were stitched for digitization, as shown in Fig. 5.

The gray-scale intensity of the stitched SEM images was digitized into numerical scale, so that the intensity of any point on the SEM image, sampled at a high pixel resolution of approximately 1.2 nm , could be represented by a numerical value. Using an edge finding algorithm, one could determine the shift in the waveguide edge with respect to a reference edge position, as shown in Fig. 5. The waveguide sidewall roughness could be determined from the root-mean-square deviation of these shifts. The autocorrelation length could also be extracted from the autocorrelation function.

To extract the autocorrelation length, we have evaluated both exponential and Gaussian autocorrelation function, described by the following:

$$\text{Exponential: } R(u) = \sigma^2 \exp\left(-\frac{|u|}{L_c}\right) \quad (3)$$

$$\text{Gaussian: } R(u) = \sigma^2 \exp\left(-\frac{u^2}{L_c^2}\right) \quad (4)$$

where u is the coordinate in the axial direction of the waveguide. We first performed a Fourier transform on the profile of the shifts and multiplied it by its complex conjugate to obtain the spectral density function $\Re(\Omega)$. The autocorrelation function $R(u)$ is the inverse Fourier transform of this spectral density function, following the relationship given by:

$$\Re(\Omega) = \int_{-\infty}^{\infty} R(u) \exp(i\Omega u) du. \quad (5)$$

The correlation length could be found from the best fit to the inverse Fourier transform using either the exponential or Gaussian autocorrelation function.

Optical Scattering Loss Measurement: Loss measurements on star coupler were performed using the setup shown in Fig. 6. Broadband light ($1530\text{--}1560 \text{ nm}$) from an Erbium-doped fiber source was coupled into the input waveguides of the SOI star coupler and the near field image of the full output waveguide array was captured by an IR camera. The image was converted to a normalized intensity profile, as shown in Fig. 6, and the relative scattering loss (excess loss) of each even waveguides was extracted from the ratio of its output intensity to the average intensity of its two adjacent reference waveguides. A polarization rotator was used so that the loss measurement of both transverse-electric (TE) and transverse-magnetic (TM) polarization could be performed, separately.

IV. RESULTS AND DISCUSSION

The r.m.s roughness and autocorrelation length extracted from SEM analysis are summarized in Table I.

The waveguides fabricated by e-beam lithography and Cr hardmask process yield the smoothest r.m.s sidewall roughness of approximately 3.6 nm . To verify the accuracy of the SEM analysis, sidewall roughness of an SOI waveguide patterned by e-beam lithography with negative resist was also characterized using atomic force microscopy (AFM). For this measurement, the sample was cleaved in close proximity to a waveguide sidewall and then inserted into the AFM sideways, (i.e., with the sidewall facing up) to be scanned with a standard tip from above. From the scanned images, we found an r.m.s roughness of 2.8 nm on the waveguide, reasonably close to the value measured using SEM.

The sidewall roughness of waveguides patterned by e-beam lithography and lift-off process is 20% larger. Among all, contact lithography process gives the largest sidewall roughness of 7.7 nm , approximately twice the roughness of the Cr hardmask process. In addition, the correlation length from contact lithography is the smallest among all, and the autocorrelation function is approximately Gaussian.

Fig. 7(a) and (b) show the near-field IR image and the measured output intensity spectrum of an output waveguide array in TE and TM polarization respectively, from a sample patterned by e-beam and dry etched with a Cr hardmask. Similar intensity spectra were obtained for waveguides fabricated by the other two processes. The intensity profile of the reference waveguides

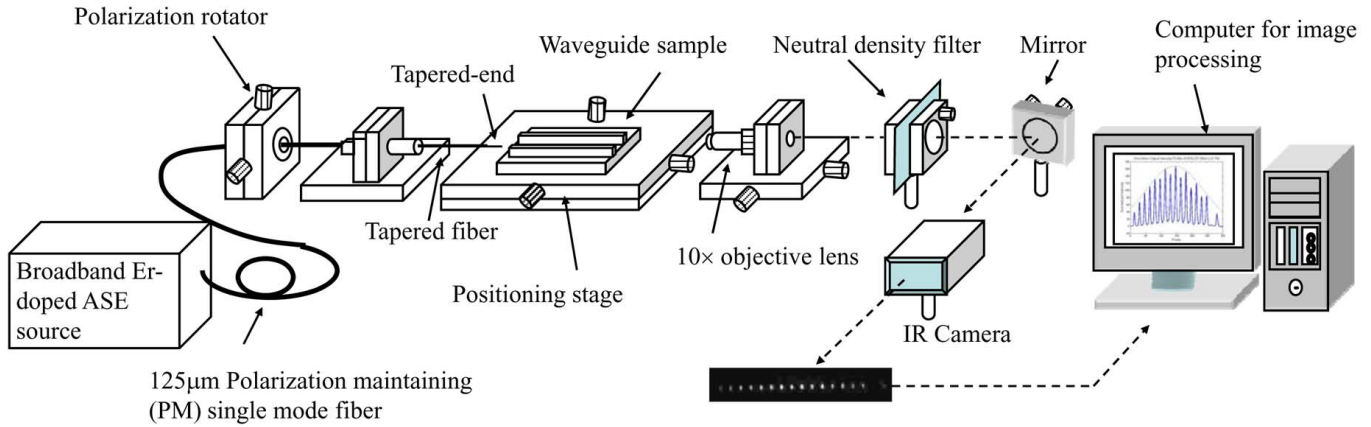


Fig. 6. Schematic of the optical scattering loss measurement using star coupler structures.

TABLE I
MEASURED R.M.S SIDEWALL ROUGHNESS AND AUTOCORRELATION LENGTH
OF SOI WAVEGUIDES FABRICATED BY THREE DIFFERENT PROCESSES

Fabrication Process	r.m.s roughness (nm)	Autocorrelation length (nm)	Autocorrelation function
Process I (Contact lithography)	7.7 ± 1.0	160 ± 25	Gaussian
Process II (E-beam lithography with positive resist)	4.3 ± 0.2	180 ± 14	Exponential
Process III (E-beam lithography with negative resist)	3.6 ± 0.2	225 ± 30	Exponential

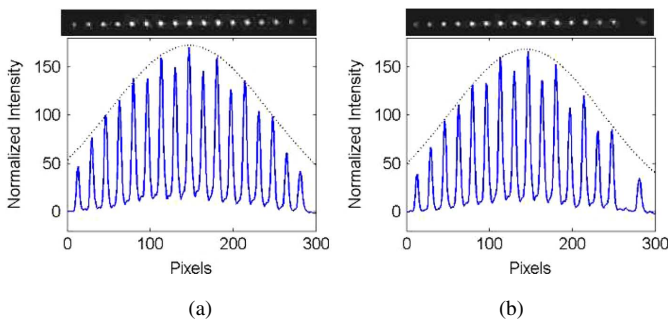


Fig. 7. Measured output intensity of waveguide array fabricated using Process III (e-beam, Cr hardmask) with the input light polarized along (a) TE direction and (b) TM direction. The horizontal axis represents the locations of the waveguides, while the vertical axis shows the measured output intensity of each waveguide. From left to right, the width of even waveguide is 1.75, 1.5, 1.25, 1, 0.8, 0.6, 0.4 and 0.2 μm , respectively. Intensities of odd waveguides follow a Gaussian envelope.

follows a far-field radiation pattern that can be approximated by a symmetric Gaussian envelope, as expected.

Fig. 8 is the plot of the measured TE and TM excess loss of the waveguides fabricated by the three different processes versus different waveguide width. The data points represent the measured experimental losses, where each data point is an average of the measured losses of three samples from the same process. The spread in the measured losses is shown by the error bars. The solid lines in Fig. 8 are the theoretical losses calculated using a modification of the Payne-Lacey model to account for varying ridge heights, as explained below. The correlation length L_c and

roughness σ used to calculate these curves were obtained by fitting the modified Payne and Lacey equation to the TM loss data in Fig. 8(b).

The excess loss from all three processes increases monotonically with decreasing width for waveguides above 1 μm . We do not have any result for waveguides smaller than 1 μm for the first two processes. This is because our contact lithography has a resolution limit of 1 μm , and the e-beam lift-off process did not produce reproducible results at waveguide widths less than 1 μm . When the width reduces to below 1 μm , the guided mode extends into the underlying slab of the ridge waveguide, and the loss becomes affected not only by scattering from the sidewalls but also from the surface roughness of the slab and leakage to the substrate [9]. We will first discuss excess losses for waveguide widths larger than 1 μm .

Samples patterned by contact lithography exhibit the highest excess losses among the three processes. Losses measured from e-beam patterned, lift-off processed waveguides are from 1 to 4 dB/cm lower than the contact lithography, and the difference becomes increasingly significant with reducing width because the effect of roughness induced scattering increases rapidly with decreasing waveguide width [7]. The Cr hardmask process with the smoothest sidewall produces the lowest excess losses among all, ranging from 0.1 to 0.5 dB/cm. The trend in the measured scattering losses from our star couplers shows very good correlation with the trend of measured roughness by SEM.

To assess the usefulness of our SOI star couplers for measuring waveguide loss, we have correlated the roughness with the measured loss. We have adapted the Payne-Lacey model [7] for loss calculation, but have introduced modifications to acquire a more accurate roughness-loss correlation. The original Payne-Lacey roughness-loss expression applies only for two-dimensional slab waveguides and the scattering loss is calculated using the core index (refractive index of the slab). Replacing the core index with an effective index will account for the change in modal propagation constant [14] in going from a slab to channel or ridge waveguide.

Since the Payne-Lacey model was developed using a simple three-layer slab waveguide model, this model should overestimate the scattering loss from etched sidewalls of a ridge waveguide. To apply the model to our SOI ridge waveguides, we de-

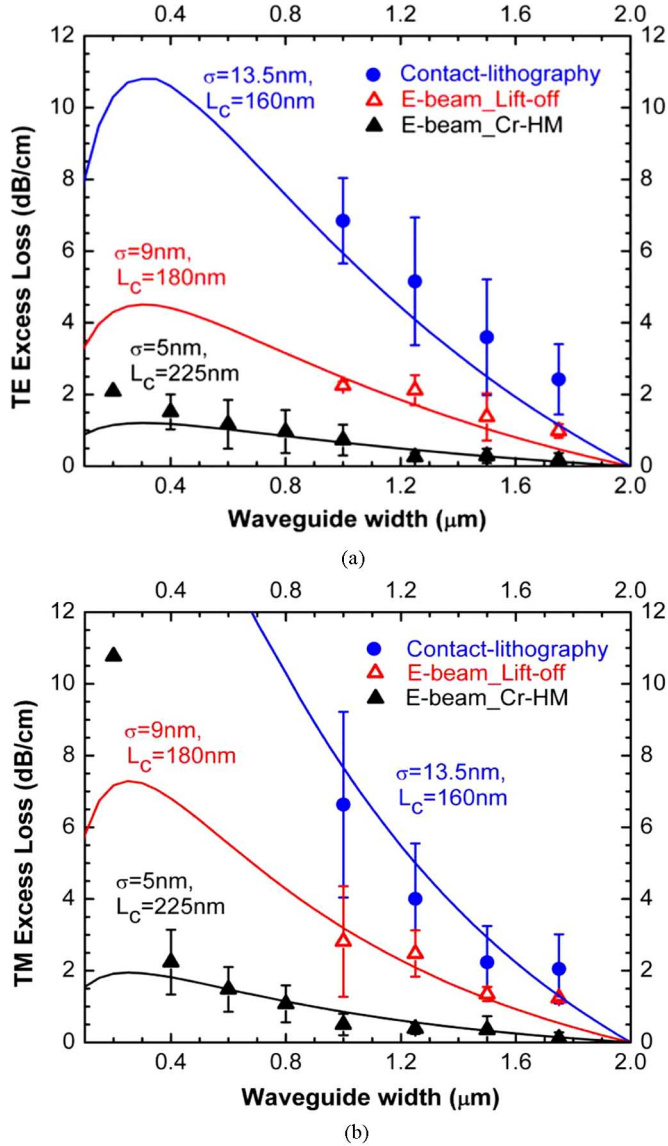


Fig. 8. Excess loss versus width of waveguides fabricated by three different processes for (a) TE and (b) TM polarization. Data points represent experimental loss while solid lines show theoretical loss. The theoretical loss is calculated by multiplying Payne-Lacey model with a scaling factor, to estimate the root-mean-square roughness.

velop a correction to modify the Payne-Lacey expression to take into account varying ridge etch depths. The loss α due to scattering depends on the modulus square of the mode electric field EE^* at the location of the scattering defect, i.e., at the rough sidewall [6], [7]. Let us assume there is a hypothetical local refractive index perturbation Δn at the waveguide surface in question. The variational theory for waveguides [15] predicts that the change in effective index δN_{eff} induced by a Δn will also vary with the overlap integral of EE^* with Δn at the perturbed surface. This suggests that δN_{eff} and the scattering loss α should both scale identically with local mode intensity EE^* overlap integral at the surface. We can therefore obtain a relative measure of the mode overlap with the ridge sidewall by using a mode solver [16] to calculate the differential change in effective index δN_{eff} induced by a very small change in waveguide ridge width δd . Taking the ratio of δN_{eff} between a ridge waveguide and

a channel waveguide (i.e., the sidewalls are completely etched down to the buried oxide layer), we obtain a scaling factor

$$s = \left(\frac{\delta N_{\text{effr}} / \delta d}{\delta N_{\text{effc}} / \delta d} \right) \quad (6)$$

where δN_{effr} is the change in the effective index in a ridge waveguide, and δN_{effc} is the change in effective index in a channel waveguide of the same width. This ratio should give an accurate estimate of the relative strength of the waveguide mode coupling to the waveguide ridge sidewalls for different etch depths. Applying these corrections to the basic Payne-Lacey expression from (1), we arrive at an expression for the sidewall scattering loss α in a ridge waveguide

$$\alpha = 4.34 \frac{\sigma^2}{\sqrt{2k_0 d^4 N_{\text{eff}}}} g \cdot f \cdot s \quad (7)$$

where s is the scaling factor of (6) and N_{eff} is the calculated effective index of the fundamental waveguide mode. As the etch depth is increased, the ridge waveguide will approach a channel waveguide geometry and the scaling factor s will approach unity. Conversely as the ridge etch becomes shallower, s approaches zero and sidewall scattering vanishes. The scaling factors of the TE and TM polarization have a slightly different dependence on waveguide width, because of the difference in the rate of change of mode index in both polarizations.

We have generated a series of excess loss contour maps with respect to roughness and autocorrelation length using the modified Payne and Lacey model of (7). The calculated loss is in TM polarization, because light is assumed to be polarized with the electric field parallel to the scattering surface in the original Payne-Lacey model [7]. The excess loss contour map of our 1.5 μm and 1 μm -wide SOI waveguides is plotted in Fig. 9(a) and (b), respectively.

The excess scattering losses on the contour maps are relative to the scattering loss of the 2 μm nominal waveguides. For any given correlation length L_c larger than 75 nm, the loss increases monotonically with roughness. As L_c increases, the dependence of loss on roughness $d\alpha/d\sigma$ reduces, as shown by the widening gap between two contour lines. However, at the vicinity of 75 nm, loss increases more rapidly with the change in roughness. When L_c decreases below 75 nm, loss decreases rapidly with decreasing L_c , and the $d\alpha/d\sigma$ again reduces.

Keeping the correlation length away from the high loss-increment region (i.e., around 75 nm for our waveguide configuration) has therefore a more prominent effect on loss reduction than merely reducing roughness. This general trend is also observed and reported in the literature [4], [14]. This indicates that loss reduction is viable through optimization of the patterning process that changes the characteristics of correlation length. In particular, we have shown that it is important to choose an optimal combination of lithography technique and resist chemistry. For instance, we have noticed that our contact lithography produces smaller L_c (~ 160 nm) with the characteristic of a Gaussian autocorrelation function, while e-beam lithography produces larger L_c (~ 180 to 225 nm) with the characteristic of an exponential autocorrelation function.

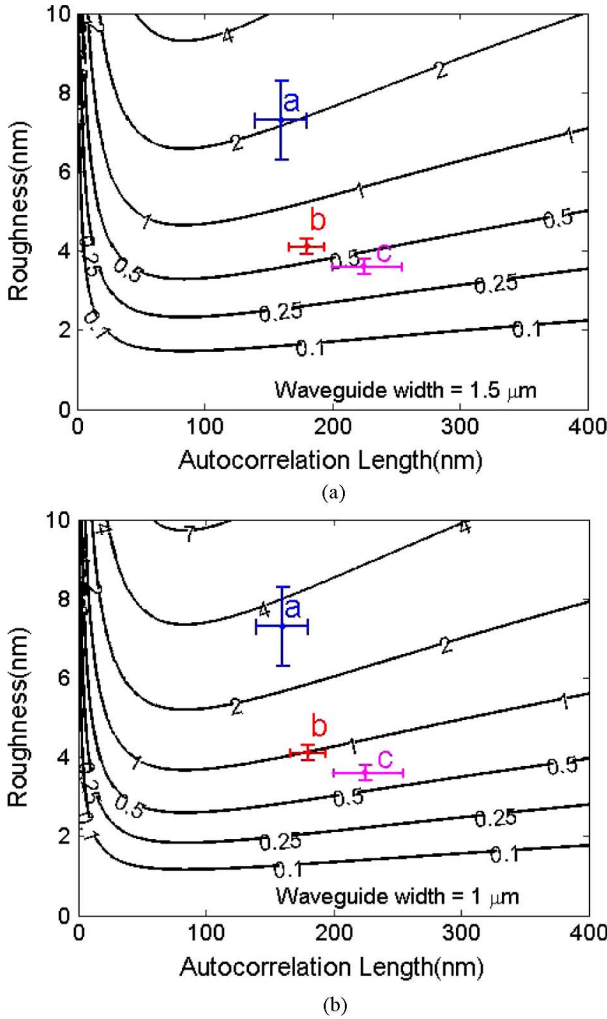


Fig. 9. Contour map of the relative TM scattering loss (in dB/cm) as a function of roughness and correlation length for waveguides of (a) 1.5 μm and (b) 1 μm width. Each solid line represents a contour of constant loss. The indicators on the map mark the measured roughness and correlation lengths with error bars obtained from the SEM analysis, for the three processes, a: Process I: contact lithography, b: Process II: e-beam lithography with positive resist and lift off, and c: Process III: e-beam lithography with negative resist.

In order to compare the measured loss with the measured σ and L_c , we have marked the measured σ and L_c from SEM with the error bars, for all three processes on the loss contour maps, as indicated by marker a, b, and c in Fig. 9(a) and (b). From the locations of the markers, we can derive the approximate values of the losses using the loss contour lines. For example, for a 1.5 μm waveguide fabricated by contact lithography process, the measured loss is around 2.24 dB/cm (TM), while the derived loss centers around 2.08 dB/cm on the contour map. Similar comparisons have been made on waveguides with 1.75, 1.5, and 1.25 μm for all three processes using the contour maps. The results of all comparisons are summarized in Table II.

In Table II, the losses calculated from the measured roughness profiles are consistently smaller than the measured waveguide losses by a factor of two or less. Similarly, the roughness values obtained by simply fitting the Payne and Lacey model to the measured loss in Fig. 8 are correspondingly larger than the measured roughness data from the SEM analysis. These differences are to be expected, since the model includes only loss

TABLE II
MEASURED SCATTERING LOSS VERSUS DERIVED LOSS FROM THE
CONTOUR MAPS (BOTH IN TM POLARIZATION)

Fabrication Process	Waveguide Width (μm)	Measured Loss (dB/cm)	Derived Loss (dB/cm)
Process I (Contact lithography)	1	6.63 ± 2.59	3.33 ± 1.26
	1.25	4.01 ± 1.54	2.79 ± 1.06
	1.5	2.24 ± 1.01	2.08 ± 0.79
	1.75	2.05 ± 0.96	1.16 ± 0.44
Process II (E-beam lithography with positive resist)	1	2.82 ± 1.54	0.99 ± 0.14
	1.25	2.48 ± 0.68	0.83 ± 0.12
	1.5	1.35 ± 0.20	0.61 ± 0.09
	1.75	1.24 ± 0.05	0.34 ± 0.06
Process III (E-beam lithography with negative resist)	1	0.50 ± 0.30	0.66 ± 0.13
	1.25	0.39 ± 0.15	0.53 ± 0.13
	1.5	0.35 ± 0.39	0.41 ± 0.08
	1.75	0.11 ± 0.17	0.23 ± 0.05

from the etched sidewalls and neglects scattering at other waveguide surfaces. Nevertheless, the results are in good qualitative agreement given the experimental uncertainties. This comparison shows the functionality of our star couplers in collecting loss data that are verifiable by process monitoring tool such as SEM and AFM. The star coupler measurement technique therefore provides useful process optimization feedback.

The measured losses reported so far are excess loss relative to the 2 μm reference waveguides. To determine the absolute scattering losses, we can use the Fabry-Perot method [9] to measure the loss of a 2 μm wide reference waveguide. By this method, we obtain losses of 7.01 dB/cm and 1.02 dB/cm for reference waveguides fabricated by contact lithography (Process I) and e-beam lithography (Process III), respectively. The absolute losses of all other smaller waveguides can then be calculated readily by adding their own excess loss to the respective 2 μm reference waveguide loss.

Next, we will discuss the excess loss of waveguides with a width of 1 μm and below. In a micron-size or sub-micron size SOI ridge waveguide, the major loss mechanisms include scattering not only from sidewall roughness, but also from slab surface roughness and the ridge lower corners. The mechanisms responsible for the scattering loss depend on the profile and confinement of the guided mode in the waveguide. Using mode matching method [16] to simulate the profile of the fundamental TE and TM modes in our waveguides, we have observed the change in mode confinement and the mode extension into the underlying slab, as the waveguide width is reduced below 1 μm .

As shown in Figs. 10 and 11, for wider ridges, the scattering loss of our waveguides (etch depth of 1.8 μm) is caused primarily by the interaction of the mode with the sidewall roughness. However, with narrowing ridge width the mode is forced into the underlying slab, the sidewall scattering decreases and loss becomes more dependent on slab surface roughness and the properties of the ridge lower corners (texture, stress distribution etc.).

The theoretical excess losses of our waveguides can be calculated from (7). In this calculation, the correlation lengths are fixed at their measured mean values as given in Table I (since the change in loss is relatively insensitive to the correlation lengths within the measured range), therefore scattering losses vary only

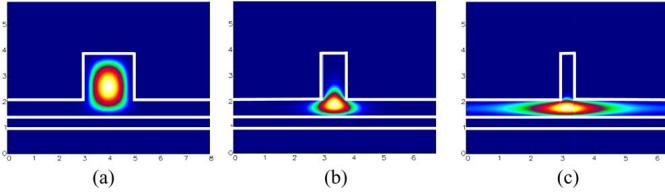


Fig. 10. TE mode profile of our SOI ridge waveguides of different widths: (a) $2\ \mu\text{m}$, (b) $0.8\ \mu\text{m}$, and (c) $0.4\ \mu\text{m}$.

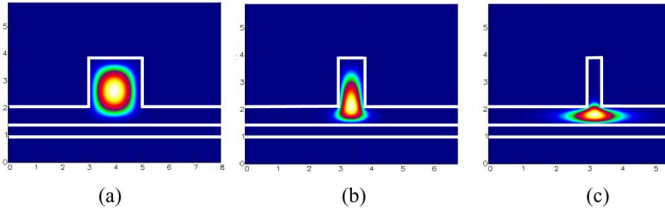


Fig. 11. TM mode profile of our SOI ridge waveguides of different widths: (a) $2\ \mu\text{m}$, (b) $0.8\ \mu\text{m}$, and (c) $0.4\ \mu\text{m}$.

TABLE III
R.M.S. ROUGHNESS MEASURED FROM SEM VERSUS ROUGHNESS
DERIVED FROM FITTING THE MEASURED LOSS

Fabrication Process	Measured roughness (nm)	Fitted roughness (nm)
Process I (Contact lithography)	7.7 ± 1.0	13.5 ± 1.5
Process II (E-beam litho, positive resist)	4.3 ± 0.2	9.0 ± 1.5
Process III (E-beam litho, negative resist)	3.6 ± 0.2	5.0 ± 1.0

with waveguide width and roughness. Various loss curves were generated using r.m.s roughness as an adjustable parameter, and the best fit to the measured loss data is shown in Fig. 8. The generated loss curves are shown by the solid lines, while the results of the fitting are summarized in Table III.

The fitted roughness agrees with the measured roughness from SEM, within a factor of 2. The remaining discrepancy may have several possible origins. The effect of surface roughness of the slab on the scattering loss of our small-dimension ridge waveguides is not accounted in the theoretical loss calculation. Therefore our theoretical loss calculated from the modified Payne-Lacey expression is underestimated, and results in a corresponding overestimate in the roughness extracted by fitting to the Payne and Lacey model. In addition, the roughness in the theoretical model is described by either an exponential or Gaussian autocorrelation function, which is at best an oversimplified approximation of the real roughness profile.

The polarization dependent losses (PDL) of the SOI samples with the roughest and smoothest sidewalls are plotted in Fig. 12.

For waveguides with rough sidewalls, such as those processed by contact lithography, when the waveguide width is large (i.e., $1.5\ \mu\text{m}$ and above), the TE modes experience more loss than the TM modes, giving rise to a positive PDL. This is because the TE mode field vector is polarized normal to the sidewalls; therefore the optical field has a higher sensitivity to sidewall roughness induced scattering [17]. The TM mode field vector, on the

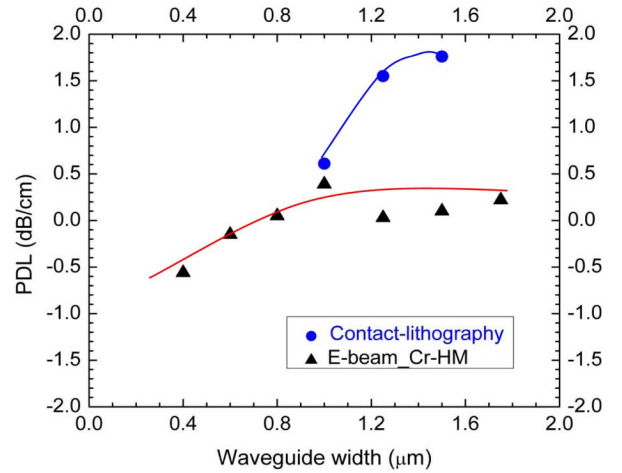


Fig. 12. Polarization dependent loss (PDL) of the waveguides as a function of waveguide width.

other hand, is polarized parallel to the sidewall, therefore is more susceptible to scattering by the slab surface roughness. As expected, the polarization dependence at large waveguide width is more prominent in samples with rough sidewalls, and less significant in the e-beam patterned waveguides with smoother sidewalls.

With decreasing waveguide width, the mode extends into the underlying slab, resulting in a change in the intensity of field overlap with the sidewall. As shown in Figs. 10 and 11, TE modes extend faster into the slab than TM modes as the waveguide width reduces. The overlapping optical field of the TE mode at the sidewall thus becomes smaller than the TM mode at the same waveguide width, for instance, at $0.8\ \mu\text{m}$. As a result, the TM mode experiences more scattering than TE mode and the PDL decreases rapidly towards negative values. The further extension of the mode profile into the underlying slab causes a change in the loss mechanism, from strongly sidewall roughness dependent to slab surface roughness dependent. This further reduces the PDL because TM modes which are more sensitive to surface scattering, will continue to experience higher loss than TE modes.

The star coupler loss measurement technique allows determination of PDL at various SOI waveguides width for a given set of roughness statistics, therefore provides useful feedback to the waveguide design and fabrication processes to minimize PDL.

V. CONCLUSION

We successfully demonstrate the effectiveness of star couplers as a test structure for assessing optical scattering loss induced by sidewall roughness. A large number of optical loss data can be obtained rapidly, since the relative loss of many waveguides can be collected in a single measurement. This provides a powerful tool for process engineers to assess various waveguide fabrication processes.

Relating the scattering loss and roughness analysis, we show that the process utilizing e-beam lithography with negative resist and a Cr hardmask for ICP dry etching produces the smoothest waveguide sidewalls and lowest scattering loss. We have modeled the measured ridge waveguide losses at various widths and

demonstrated that the fitted sidewall roughness is in reasonable agreement with the measured roughness from SEM. In conclusion, this star coupler technique is capable of studying roughness induced scattering loss and will enable optimization of process parameters that critically affect waveguide sidewall roughness.

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Boris Lamontagne, photograph and biography not available at the time of publication.

Jens H. Schmid, photograph and biography not available at the time of publication.

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